

Sem 2 AY22-23 MH1101 Calculus II Test (1 hour)

Name:

Matriculation Number:

Tutorial Group:

Question 1 [16 marks]

(a) Write the following limit as a definite integral (do not evaluate it).

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3i}{n^2 + i^2}$$

(b) If $f(x) = \int_{2x}^{x^2} \cos(t^2) dt$, find the value of $f'(1)$. You may leave your final answer in terms of cosine.

Answer.

More space for answers.

Question 2 [16 marks]

- (a) Let R be the region in the first quadrant bounded by the curves $y = x^2$, $x + y = 30$ and the x -axis (i.e. the horizontal line $y = 0$). Compute the volume of the solid formed by revolving R about the vertical line $x = 30$.
- (b) Determine if the following improper integral converges, and if so, evaluate it. Justify your answer.

$$\int_0^{\infty} \frac{1}{(x^2 + 1)^{3/2}} dx.$$

Answer.

More space for answers.

More space for answers.

Question 3 [18 marks]

(a) Evaluate the following integrals:

(i) $\int 4 \sin^3(4x) \cos^6(4x) dx$

(ii) $\int (\ln(x+1))^2 dx$ Hint: You may consider a substitution.

(b) Assume that f is a function with a continuous second derivative. Show that $f(b) = f(a) + f'(a)(b-a) + \int_a^b f''(x)(b-x) dx$. Hence or otherwise, show that $|f(b) - f(a) - f'(a)(b-a)| \leq \frac{b^2}{2}$ for any positive number b .

Answer.

More space for answers.

More space for answers.

More space for answers.

Formulae Table

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int e^x dx = e^x + C$$

$$\int k dx = kx + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \tan x dx = \ln |\sec x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1} x$$

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

$$\sin 2x = 2 \sin x \cos x$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\sin A \cos B = \frac{1}{2} (\sin(A - B) + \sin(A + B))$$

$$\sin A \sin B = \frac{1}{2} (\cos(A - B) - \cos(A + B))$$

$$\cos A \cos B = \frac{1}{2} (\cos(A - B) + \cos(A + B))$$

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

– End of Test –